TEST PLAN DOCUMENT

Student Record System

FR-001 — Record and Update Attendance

Requirement: Authorized faculty can record attendance for a selected subject/date and update existing attendance.

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
ST_001_01	Record and Update Attendance	Record attendance with all students marked	Faculty is logged in; subject and enrolled students exist.	1. Open Manage Attendance. 2. Select subject and date. 3. Mark every student Present/Absent. 4. Submit.	Subject: CSE101; date: 2026-10-12; Student A: Present; Student B: Absent	Attendance is saved and appears in the selected subject/date records.		Pending
ST_001_02	Record and Update Attendance	Reject submission when a student has no status	Faculty is logged in; enrolled students exist.	1. Open attendance entry. 2. Leave one student's status blank. 3. Submit.	Student A: Present; Student B: blank	Submission is blocked and the system asks faculty to complete attendance.		Pending
IT_001_03	Record and Update Attendance	Update an existing attendance record	An attendance record already exists; faculty is authorized.	1. Find the record by subject/date. 2. Change one student's status. 3. Save.	Student A: Absent → Present	Updated status is saved and shown in attendance details.		Pending
IT_001_04	Record and Update Attendance	Preserve attendance for other students when one is updated	Attendance exists for multiple students.	1. Open existing attendance. 2. Change one student's status. 3. Save. 4. Review all students.	A: Absent → Present; B: Present	Only the selected student's status changes; other records remain unchanged.		Pending
ST_001_05	Record and Update Attendance	Prevent unauthorized attendance editing	A student account is available; attendance exists.	1. Log in as student. 2. Try to open attendance management or submit an update.	Student account; existing attendance record	Access is denied and no attendance data is modified.		Pending
IT_001_06	Record and Update Attendance	Reject duplicate attendance submission for same subject/date if duplicates are disallowed	An attendance record already exists for the same subject/date.	1. Open attendance entry for the same subject/date. 2. Submit a second record without choosing update.	Subject CSE101; same date as existing record	System prevents duplicate records or clearly routes faculty to update the existing record, according to the implemented rule.		Pending
IT_001_07	Record and Update Attendance	Handle save failure without falsely confirming success	Test environment can simulate a save/database failure.	1. Enter valid attendance. 2. Trigger save failure. 3. Observe response.	Valid attendance; simulated database unavailable	System displays a failure message and does not show a false success confirmation.		Pending
ST_001_08	Record and Update Attendance	Verify attendance persists after reopening	Faculty has saved an attendance record.	1. Save attendance. 2. Navigate away. 3. Reopen the same subject/date.	Subject CSE101; date 2026-10-12	Saved attendance values remain correct after reopening.		Pending

FR-002 — Calculate Attendance Percentage

Requirement: Calculate each student's attendance percentage for a subject.

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
UT_002_01	Calculate Attendance Percentage	Calculate attendance percentage for a typical case	Attendance records exist for the student and subject.	1. Open attendance summary. 2. View percentage.	Attended 8 classes; total 10	Displayed percentage is 80%.		Pending
UT_002_02	Calculate Attendance Percentage	Calculate 100 percent attendance	Attendance records exist.	1. View attendance percentage.	Attended 10; total 10	Displayed percentage is 100%.		Pending
UT_002_03	Calculate Attendance Percentage	Calculate zero percent attendance	Total classes exist and student attended none.	1. View attendance percentage.	Attended 0; total 10	Displayed percentage is 0%.		Pending
UT_002_04	Calculate Attendance Percentage	Handle no classes conducted	Subject has no recorded classes.	1. Open attendance percentage view.	Attended 0; total 0	System avoids division by zero and displays a clear unavailable/not-yet-calculated message.		Pending
IT_002_05	Calculate Attendance Percentage	Recalculate after attendance is updated	Student attendance has been calculated; a record can be changed.	1. Change one attendance status. 2. Reopen percentage.	Previously 8/10; update one Absent to Present so result becomes 9/10	Percentage updates to 90%.		Pending
IT_002_06	Calculate Attendance Percentage	Calculate percentages independently per subject	Student has records in two subjects.	1. Open each subject's attendance summary.	CSE101: 8/10; MAT101: 6/8	CSE101 shows 80%; MAT101 shows 75%; values are not mixed.		Pending
ST_002_07	Calculate Attendance Percentage	Show message when attendance data is missing	Student/subject has no attendance records.	1. Open attendance percentage view.	No records for PHY101	System indicates attendance data is unavailable instead of showing a misleading value.		Pending
UT_002_08	Calculate Attendance Percentage	Verify percentage rounding/display rule	Attendance records support a fractional percentage.	1. Open attendance percentage view.	Attended 2; total 3	System displays approximately 66.67% or the documented rounding format consistently.		Pending

FR-003 — Enter and Update Marks

Requirement: Faculty can enter and update student marks.

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
ST_003_01	Enter and Update Marks	Enter marks for a student	Faculty is logged in; student and subject exist.	1. Open marks entry. 2. Select subject/student. 3. Enter marks. 4. Save.	CSE101; Student A; 78/100	Marks are saved and appear in the student's academic record.		Pending
IT_003_02	Enter and Update Marks	Update existing marks	Marks already exist for a student and subject.	1. Open marks entry. 2. Change marks. 3. Save.	Previous 65/100; new 82/100	New marks are saved and displayed.		Pending
IT_003_03	Enter and Update Marks	Retrieve existing marks before editing	Existing marks are present.	1. Select the student and subject in marks entry.	Student A; CSE101; existing marks 65	System displays the existing value so faculty can review/update it.		Pending
ST_003_04	Enter and Update Marks	Reject marks above maximum	Faculty is logged in; maximum marks are configured.	1. Enter marks above maximum. 2. Submit.	101/100	System rejects the entry and displays an appropriate validation message.		Pending
UT_003_05	Enter and Update Marks	Reject negative marks	Faculty is logged in; marks range is configured.	1. Enter negative marks. 2. Submit.	-5/100	System rejects the entry and does not save the invalid marks.		Pending
ST_003_06	Enter and Update Marks	Prevent unauthorized marks editing	Student account is available; marks exist.	1. Log in as student. 2. Attempt to open marks editing or submit changes.	Student account; Student A's marks	Access is denied and marks remain unchanged.		Pending
ST_003_07	Enter and Update Marks	Handle missing marks input	Faculty is on the marks-entry screen.	1. Leave marks blank. 2. Submit.	Blank marks field	System requests valid marks and does not save an unintended value.		Pending
IT_003_08	Enter and Update Marks	Verify marks persist after reopening	Marks have been saved successfully.	1. Navigate away. 2. Reopen the same student/subject record.	Student A; CSE101; 82/100	The saved marks are still 82/100.		Pending

FR-004 — Calculate Semester GPA

Requirement: Convert marks to grade points using institutional rules and calculate semester GPA using subject credits.

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
UT_004_01	Calculate Semester GPA	Calculate GPA for multiple subjects	Marks, grade mapping, and credits exist for the semester.	1. Open GPA view/calculation. 2. Review semester GPA.	Example: grade points 8 and 9; credits 3 and 4; weighted GPA=(8×3+9×4)/7	GPA is calculated using the configured weighted-credit formula (8.57 for this example, subject to rounding rule).		Pending
UT_004_02	Calculate Semester GPA	Convert marks to grade points at a grading boundary	Institution grading rules are configured.	1. Enter/view marks at a grade boundary. 2. Calculate GPA.	Use the exact configured boundary, e.g. 90/100	Grade point matches the institution's configured mapping for that boundary.		Pending
UT_004_03	Calculate Semester GPA	Convert marks to grade points below a boundary	Institution grading rules are configured.	1. Enter/view marks just below a boundary. 2. Calculate GPA.	Use one mark below a configured boundary, e.g. 89/100	Grade point follows the configured lower grade band.		Pending
IT_004_04	Calculate Semester GPA	Handle missing marks	At least one subject has no marks.	1. Request GPA calculation for the semester.	CSE101 marks available; MAT101 marks missing	System indicates required marks are missing and does not present a misleading final GPA.		Pending
IT_004_05	Calculate Semester GPA	Handle missing subject credits	A subject has marks but no credit value.	1. Request GPA calculation.	CSE101 marks 78; credit value missing	System indicates missing credit information and does not calculate an incorrect GPA.		Pending
IT_004_06	Calculate Semester GPA	Recalculate GPA after marks update	A semester GPA exists and marks can be updated.	1. Update a subject's marks. 2. Reopen GPA.	Change one subject's marks to a different grade band	GPA is recalculated based on updated grade points and credits.		Pending
ST_004_07	Calculate Semester GPA	Display GPA for the selected semester	A student has results in multiple semesters.	1. Select one semester. 2. View GPA.	Semester 1 and Semester 2 have different marks	Displayed GPA corresponds to the selected semester only.		Pending
UT_004_08	Calculate Semester GPA	Handle invalid grade/credit configuration	Test data includes an invalid grade mapping or credit value.	1. Request GPA calculation.	Invalid grade mapping or non-permitted credit value	System reports invalid academic configuration and avoids displaying a misleading GPA.		Pending

FR-005 — View Academic Records by Semester

Requirement: Students can view their attendance, marks, grades, GPA, and CGPA, and select a semester.

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
ST_005_01	View Academic Records by Semester	View academic records for a selected semester	Student is logged in and has academic records.	1. Open My Records. 2. Select a semester.	Semester 1 with attendance, marks, grades and GPA	Selected semester's records are displayed with CGPA.		Pending
ST_005_02	View Academic Records by Semester	Switch between semesters	Student has records in at least two semesters.	1. Select Semester 1. 2. Select Semester 2.	Semester 1 and Semester 2 records differ	Each selection displays the matching semester's records.		Pending
ST_005_03	View Academic Records by Semester	Show no-records message for an empty semester	Student account has no records for one semester.	1. Select the empty semester.	Semester 3 with no data	System indicates that no records are available for that semester.		Pending
ST_005_04	View Academic Records by Semester	Restrict records to the logged-in student	Two student accounts have different records.	1. Log in as Student A. 2. Open My Records.	Student A and Student B test data	Only Student A's academic records are shown.		Pending
IT_005_05	View Academic Records by Semester	Display attendance, marks, grades, GPA and CGPA	Student has complete academic data.	1. Open My Records. 2. Select a semester.	Complete test record for Semester 1	All required record categories are displayed correctly.		Pending
ST_005_06	View Academic Records by Semester	Handle session expiration while viewing records	Session expiration can be simulated.	1. Log in. 2. Expire session. 3. Try to view records.	Expired session	System requires re-authentication and does not expose protected records.		Pending
IT_005_07	View Academic Records by Semester	Refresh records after marks are updated	Faculty has updated marks for the student.	1. Log in as student. 2. Open the relevant semester records.	Updated marks and resulting grade/GPA	Displayed records reflect the saved latest data according to calculation rules.		Pending
ST_005_08	View Academic Records by Semester	Handle unavailable academic data	A record source can be made unavailable in the test environment.	1. Select a semester while data retrieval fails.	Simulated retrieval failure	System displays a clear error/unavailable message rather than corrupt or unrelated records.		Pending

FR-006 — User Login and Logout

Requirement: Users can log in and log out securely.

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
ST_006_01	User Login and Logout	Login with valid student credentials	Student account exists and is active.	1. Open login page. 2. Enter valid credentials. 3. Submit.	Valid student username/password	Student is authenticated and shown the student dashboard.		Pending
ST_006_02	User Login and Logout	Login with valid faculty credentials	Faculty account exists and is active.	1. Open login page. 2. Enter valid faculty credentials. 3. Submit.	Valid faculty username/password	Faculty is authenticated and shown the faculty dashboard.		Pending
UT_006_03	User Login and Logout	Reject incorrect password	User account exists.	1. Enter valid username and wrong password. 2. Submit.	Correct username; incorrect password	Login is rejected and an appropriate error is shown.		Pending
UT_006_04	User Login and Logout	Reject unknown username	No account exists for the username.	1. Enter unknown username and any password. 2. Submit.	Username: unknown_test_user	Login is rejected without granting access.		Pending
ST_006_05	User Login and Logout	Prevent access to protected page without login	User is logged out.	1. Directly navigate to a protected records page.	Unauthenticated session	System redirects to login or denies access.		Pending
ST_006_06	User Login and Logout	Logout ends the session	User is logged in.	1. Click Logout. 2. Try to reopen a protected page using browser navigation.	Valid authenticated session	Session ends; protected content is not accessible without logging in again.		Pending
IT_006_07	User Login and Logout	Prevent role confusion after login	Student and faculty accounts exist.	1. Log in as student. 2. Attempt to open a faculty-only page.	Student account; faculty route/page	Access is denied to the student account.		Pending
UT_006_08	User Login and Logout	Handle empty credentials	Login page is open.	1. Leave username and password blank. 2. Submit.	Blank username and password	System requires the credentials and does not create a session.		Pending

FR-007 — Search Students and View Profiles

Requirement: Authorized faculty can search by student ID and view the matching profile.

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
ST_007_01	Search Students and View Profiles	Find a student using a valid ID	Faculty is logged in and student exists.	1. Open student search. 2. Enter student ID. 3. Search.	Student ID: STU1001	Matching student's profile is displayed.		Pending
ST_007_02	Search Students and View Profiles	Show not-found message for unknown ID	Faculty is logged in.	1. Search for an ID that does not exist.	Student ID: STU9999	System displays a no-matching-student message.		Pending
UT_007_03	Search Students and View Profiles	Reject empty student ID	Faculty is on search page.	1. Leave ID blank. 2. Submit search.	Blank student ID	System asks for an ID or displays a validation message.		Pending
UT_007_04	Search Students and View Profiles	Ignore surrounding spaces in student ID if supported	Faculty is logged in; test student exists.	1. Search with spaces around the ID.	' STU1001 '	System finds the student if input trimming is part of the implemented validation; otherwise displays a clear validation response.		Pending
IT_007_05	Search Students and View Profiles	Display correct profile for selected ID	At least two student records exist.	1. Search for Student B's ID. 2. Verify profile fields.	Student B ID and known profile data	Displayed profile belongs to Student B, not another student.		Pending
ST_007_06	Search Students and View Profiles	Prevent student from using faculty profile search	Student account exists.	1. Log in as student. 2. Try to open faculty student-search/profile feature.	Student credentials	Access is denied.		Pending
IT_007_07	Search Students and View Profiles	Handle duplicate/ambiguous IDs safely	Test database can be checked for uniqueness.	1. Search using an ID in test data.	Unique student ID constraint test	System returns one correct profile; IDs should be unique, and duplicates must not cause the wrong profile to be shown.		Pending
IT_007_08	Search Students and View Profiles	Handle profile retrieval failure	Faculty is logged in; retrieval failure can be simulated.	1. Search for a valid student while profile retrieval fails.	Student ID STU1001; simulated service/database failure	System displays an error and does not show incorrect profile details.		Pending

FR-008 — Maintain Subject Details and Credits

Requirement: Maintain subject details and corresponding credit values.

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
ST_008_01	Maintain Subject Details and Credits	Create a subject with valid credits	Authorized subject manager is logged in.	1. Open subject management. 2. Add subject and credit value. 3. Save.	Subject CSE101; credits 3	Subject and credit value are saved.		Pending
IT_008_02	Maintain Subject Details and Credits	Update an existing subject credit value	Subject exists and user is authorized.	1. Open subject details. 2. Change credits. 3. Save.	CSE101: credits 3 → 4	Updated credit value is saved and displayed.		Pending
UT_008_03	Maintain Subject Details and Credits	Reject missing subject name	Authorized subject manager is logged in.	1. Add subject with blank name. 2. Save.	Name blank; credits 3	System rejects the incomplete subject details.		Pending
UT_008_04	Maintain Subject Details and Credits	Reject missing credit value	Subject management form is open.	1. Enter subject name but leave credits blank. 2. Save.	Name CSE102; credits blank	System asks for a valid credit value and does not save an incomplete record.		Pending
UT_008_05	Maintain Subject Details and Credits	Reject invalid credit value	Credit validation rules are configured.	1. Enter a non-numeric or non-permitted credit value. 2. Save.	Credits: 'abc' or -1	System rejects the invalid credit value.		Pending
ST_008_06	Maintain Subject Details and Credits	Prevent unauthorized subject changes	Unauthorized student account exists.	1. Log in as student. 2. Try to open subject management or update credits.	Student account; existing subject	Access is denied and subject details remain unchanged.		Pending
IT_008_07	Maintain Subject Details and Credits	Verify updated credits are used in GPA	Subject credits and student marks exist; GPA feature is available.	1. Update a subject credit value. 2. Recalculate GPA.	CSE101 credits 3 → 4; known grade points	GPA calculation uses the updated credit value.		Pending
IT_008_08	Maintain Subject Details and Credits	Prevent duplicate subject records where subject IDs must be unique	A subject with the same ID already exists.	1. Try to add another subject using the existing ID.	Existing subject ID CSE101	System rejects the duplicate ID or prevents duplicate subject records according to the data model.		Pending

FR-009 — Calculate CGPA

Requirement: Calculate CGPA from semester results and corresponding subject credits using the institution's formula.

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
UT_009_01	Calculate CGPA	Calculate CGPA for multiple semesters	Semester GPA/grade points and credits exist for multiple semesters.	1. Open CGPA view/calculation. 2. Verify the result.	Use configured institution formula with known semester data	Displayed CGPA matches the institution's prescribed formula.		Pending
UT_009_02	Calculate CGPA	Calculate CGPA when only one semester exists	Student has complete results for one semester.	1. Request CGPA.	One semester with known results and credits	CGPA follows the institution's formula for the available semester data.		Pending
IT_009_03	Calculate CGPA	Handle missing semester results	At least one required semester result is missing.	1. Request CGPA.	Semester 1 complete; Semester 2 missing required data	System identifies incomplete data and does not display a misleading CGPA.		Pending
IT_009_04	Calculate CGPA	Handle missing subject credits	A required subject credit is missing.	1. Request CGPA.	One subject with no credit value	System reports missing credit information and avoids incorrect calculation.		Pending
IT_009_05	Calculate CGPA	Recalculate CGPA after marks/GPA changes	Existing CGPA exists; a result can be updated.	1. Update marks or semester result. 2. Reopen CGPA.	Change one result to a different grade point	CGPA is recalculated using updated results.		Pending
UT_009_06	Calculate CGPA	Verify semester records are not double-counted	Multiple semester results exist.	1. Calculate CGPA. 2. Compare with formula using each semester once.	Known results for Semester 1 and Semester 2	Each applicable result is counted according to the configured formula, without duplicate counting.		Pending
IT_009_07	Calculate CGPA	Display CGPA consistently in student records	CGPA has been calculated.	1. Open CGPA view. 2. Open My Records.	Same student and same data	CGPA shown in both places is consistent.		Pending
UT_009_08	Calculate CGPA	Handle invalid academic data during CGPA calculation	Test data includes invalid or incomplete grade/credit information.	1. Request CGPA.	Invalid grade point or credit value	System reports invalid data and does not return a misleading CGPA.		Pending

FR-010 — Filter Attendance Records

Requirement: Faculty can filter attendance records by subject and date.

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
ST_010_01	Filter Attendance Records	Filter attendance by subject and date	Attendance records exist for several subjects/dates.	1. Open attendance records. 2. Select subject and date. 3. Apply filters.	CSE101; 2026-10-12	Only records matching CSE101 and the selected date are displayed.		Pending
ST_010_02	Filter Attendance Records	Filter by a subject with several dates	Attendance exists for selected subject on multiple dates.	1. Select subject and one date. 2. Apply filter.	CSE101; date 2026-10-12	Only records for the selected date appear.		Pending
IT_010_03	Filter Attendance Records	Filter by a date across subjects	Records exist for several subjects on the same date.	1. Select date and a subject as required by the interface. 2. Apply.	Date 2026-10-12; CSE101	Displayed records obey the selected filter criteria.		Pending
ST_010_04	Filter Attendance Records	Show no-match message	No records match the chosen filters.	1. Select a subject/date with no records. 2. Apply.	PHY101; date with no records	System indicates no matching attendance records were found.		Pending
UT_010_05	Filter Attendance Records	Reject invalid date input	Filter form is open.	1. Enter/select an invalid date if free text is supported. 2. Apply.	Invalid date string or end date before start date if range is supported	System rejects invalid filter input with a clear message.		Pending
IT_010_06	Filter Attendance Records	Change filters and refresh results	Records exist for two subjects.	1. Filter CSE101. 2. Change to MAT101. 3. Apply again.	CSE101 then MAT101	Results refresh and show only the newly selected subject's records.		Pending
ST_010_07	Filter Attendance Records	Prevent unauthorized attendance browsing	Student account exists; faculty filter feature is restricted.	1. Log in as student. 2. Attempt to open faculty attendance filtering.	Student credentials	Access is denied or only the student's permitted records are available, consistent with role permissions.		Pending
IT_010_08	Filter Attendance Records	Handle retrieval failure	Attendance records service/database failure can be simulated.	1. Apply valid filters while retrieval fails.	CSE101; 2026-10-12; simulated failure	System displays an error and does not present stale results as current.		Pending

FR-011 — Validate Entered Marks

Requirement: Reject marks outside the permitted range.

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
UT_011_01	Validate Entered Marks	Accept marks at the minimum permitted value	Marks form is available; range is configured.	1. Enter minimum marks. 2. Submit.	0/100 if configured minimum is 0	System accepts and saves the boundary value.		Pending
UT_011_02	Validate Entered Marks	Accept marks at the maximum permitted value	Marks form is available; maximum is configured.	1. Enter maximum marks. 2. Submit.	100/100	System accepts and saves the boundary value.		Pending
UT_011_03	Validate Entered Marks	Reject marks below minimum	Marks form is available.	1. Enter a value below minimum. 2. Submit.	-1/100	System rejects the marks and shows a validation message.		Pending
UT_011_04	Validate Entered Marks	Reject marks above maximum	Marks form is available.	1. Enter a value above maximum. 2. Submit.	101/100	System rejects the marks and shows a validation message.		Pending
UT_011_05	Validate Entered Marks	Reject non-numeric marks	Marks form is available.	1. Enter letters in the marks field. 2. Submit.	'abc'	System rejects non-numeric input and does not save it.		Pending
UT_011_06	Validate Entered Marks	Reject missing marks	Marks form is available.	1. Leave marks blank. 2. Submit.	Blank marks	System requests a marks value and does not save an unintended value.		Pending
IT_011_07	Validate Entered Marks	Preserve previous marks after invalid update	Existing valid marks are saved.	1. Attempt to replace them with invalid marks. 2. Reopen the record.	Existing 75/100; attempted update 120/100	Invalid update is rejected and the existing valid marks remain unchanged.		Pending
IT_011_08	Validate Entered Marks	Validate different maximum marks per assessment if configured	Assessment has a configured maximum different from 100.	1. Enter marks at and above that assessment's maximum.	Quiz maximum 20; values 20 and 21	20 is accepted and 21 is rejected when the configured maximum is 20.		Pending

FR-012 — Generate Class-Wise Attendance Report

Requirement: Faculty can generate an attendance report for a selected class/subject.

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
ST_012_01	Generate Class-Wise Attendance Report	Generate report for a selected subject/class	Faculty is logged in; attendance records exist.	1. Open attendance reports. 2. Select class/subject. 3. Generate.	CSE101; class CSE-A	Report is generated and contains attendance records for the selected class/subject.		Pending
IT_012_02	Generate Class-Wise Attendance Report	Verify report contains correct students	Attendance exists for known enrolled students.	1. Generate report. 2. Compare listed students to enrollment/attendance data.	Students A, B, C enrolled in CSE101	Report contains the correct students and their attendance details.		Pending
IT_012_03	Generate Class-Wise Attendance Report	Generate report when attendance is incomplete	Some attendance records are missing.	1. Generate report for the selected class/subject.	One student has no recorded status	Report reflects available data accurately and does not invent attendance values.		Pending
ST_012_04	Generate Class-Wise Attendance Report	Show message when no attendance records exist	Selected class/subject has no attendance records.	1. Generate report.	PHY101 with no attendance records	System states that no report can be generated or no records are available.		Pending
ST_012_05	Generate Class-Wise Attendance Report	Prevent unauthorized report access	Student account exists; report generation is faculty-only.	1. Log in as student. 2. Attempt to generate class-wide report.	Student credentials	Access is denied or limited according to the defined role permissions.		Pending
IT_012_06	Generate Class-Wise Attendance Report	Verify report data after attendance update	Attendance record has been updated.	1. Update attendance. 2. Generate the report again.	Student A changes Absent → Present	Report reflects the latest saved attendance.		Pending
IT_012_07	Generate Class-Wise Attendance Report	Handle report generation failure	Report failure can be simulated.	1. Select valid class/subject. 2. Generate while report service fails.	CSE101; simulated generation failure	System displays an error and does not claim a report was generated.		Pending
ST_012_08	Generate Class-Wise Attendance Report	Verify report is limited to selected class/subject	Records exist across multiple classes/subjects.	1. Generate a report for one class/subject.	CSE-A/CSE101 while other class records exist	Report does not include records from unrelated classes or subjects.		Pending

FR-013 — Notify Students of Low Attendance

Requirement: Notify students whose attendance falls below the institution's threshold.

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
UT_013_01	Notify Students of Low Attendance	Generate notification below threshold	Student attendance and threshold are configured.	1. Trigger/check notification process. 2. Inspect student notifications.	Attendance 65%; threshold 75%	A low-attendance notification is generated for the student.		Pending
UT_013_02	Notify Students of Low Attendance	Do not notify when attendance equals threshold	Threshold and attendance are configured.	1. Run notification check.	Attendance 75%; threshold 75%	No low-attendance notification is generated if the requirement is strictly below threshold.		Pending
UT_013_03	Notify Students of Low Attendance	Do not notify when attendance exceeds threshold	Attendance is above threshold.	1. Run notification check.	Attendance 80%; threshold 75%	No low-attendance notification is generated.		Pending
IT_013_04	Notify Students of Low Attendance	Notify only affected students	Multiple students have different attendance percentages.	1. Run notification check. 2. Review notifications.	A 60%, B 80%, C 70%; threshold 75%	A and C receive notifications; B does not.		Pending
IT_013_05	Notify Students of Low Attendance	Use configured institution threshold	Threshold can be configured by authorized setup.	1. Change threshold. 2. Run notification check.	Threshold 80%; student attendance 78%	Notification decision uses the configured 80% threshold.		Pending
IT_013_06	Notify Students of Low Attendance	Avoid duplicate notification on repeated check if duplicates are disallowed	A low-attendance notification already exists; duplicate policy is defined.	1. Run the notification process twice. 2. Review notifications.	Attendance 60%; threshold 75%	System follows the defined duplicate-notification policy and does not create unintended repeated alerts.		Pending
UT_013_07	Notify Students of Low Attendance	Do not notify when attendance data is unavailable	Student has no usable attendance data.	1. Run notification check.	No classes/attendance records	System does not generate a misleading low-attendance notification and handles unavailable data clearly.		Pending
IT_013_08	Notify Students of Low Attendance	Handle notification delivery failure	Notification failure can be simulated.	1. Trigger a low-attendance notification. 2. Simulate delivery failure.	Attendance 60%; threshold 75%; simulated delivery failure	System records/reports the failure and supports retry according to the implementation.		Pending

FR-014 — View Subject-Wise Marks Summary

Requirement: Faculty can view a summary of marks for students in a selected subject.

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
ST_014_01	View Subject-Wise Marks Summary	View marks summary for a selected subject	Faculty is logged in; marks exist.	1. Open subject-wise marks summary. 2. Select subject.	CSE101 with marks for several students	System displays the marks summary for CSE101.		Pending
IT_014_02	View Subject-Wise Marks Summary	Verify summary includes the correct students and marks	Known marks exist for enrolled students.	1. Generate summary. 2. Compare values with stored marks.	A: 78; B: 84; C: 69	Summary lists the correct students and their marks.		Pending
ST_014_03	View Subject-Wise Marks Summary	Show no-marks message	Selected subject has no marks entered.	1. Select subject. 2. Open summary.	PHY101; no marks	System indicates that marks are not available.		Pending
ST_014_04	View Subject-Wise Marks Summary	Keep summary limited to selected subject	Marks exist for multiple subjects.	1. Select CSE101. 2. View summary.	CSE101 and MAT101 both have data	Only CSE101 marks are included.		Pending
IT_014_05	View Subject-Wise Marks Summary	Reflect updated marks in summary	A student's marks have been updated.	1. Update marks. 2. Reopen subject summary.	Student A: 65 → 82	Summary shows the latest saved marks.		Pending
ST_014_06	View Subject-Wise Marks Summary	Prevent unauthorized access to class marks	Student account exists; summary is faculty-only.	1. Log in as student. 2. Attempt to open subject-wise summary.	Student credentials	Access is denied or restricted according to role permissions.		Pending
IT_014_07	View Subject-Wise Marks Summary	Handle marks retrieval failure	Database/retrieval failure can be simulated.	1. Open summary during retrieval failure.	CSE101; simulated failure	System displays an error and does not show misleading data.		Pending
IT_014_08	View Subject-Wise Marks Summary	Verify summary handles students with missing marks	Some enrolled students have no marks entered.	1. Generate summary for the subject.	Student A: 78; Student B: no marks	System clearly represents missing marks (e.g. blank/not entered) and does not treat them as zero unless specified by the project rules.		Pending

FR-015 — Download Academic Records

Requirement: Students can download their academic records as a report.

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
ST_015_01	Download Academic Records	Download academic report with complete records	Student is logged in and has academic records.	1. Open Download Academic Records. 2. Generate report. 3. Download it.	Student A; attendance, marks, grades, GPA, CGPA	A report file is made available and downloaded successfully with the student's available academic records.		Pending
IT_015_02	Download Academic Records	Verify downloaded report content	Student has known academic data.	1. Download report. 2. Open the downloaded file. 3. Compare values.	Known Student A record for Semester 1	Downloaded content matches the student's stored academic records.		Pending
ST_015_03	Download Academic Records	Handle student with no academic records	Student account has no academic records.	1. Request a download.	No academic records	System informs the student that a report cannot be generated or records are unavailable.		Pending
ST_015_04	Download Academic Records	Ensure report belongs only to logged-in student	Two student accounts have different records.	1. Log in as Student A. 2. Download report. 3. Inspect it.	Student A and Student B have different data	Report contains only Student A's authorized records.		Pending
IT_015_05	Download Academic Records	Download report for selected semester if selection is offered	Student has records in multiple semesters and report flow supports semester selection.	1. Select a semester. 2. Generate and download report.	Semester 1 selected	Report includes the selected semester's records and any other fields required by the report design.		Pending
IT_015_06	Download Academic Records	Handle report generation failure	Report generation failure can be simulated.	1. Request report while generation fails.	Student A; simulated generation failure	System displays an error and does not claim that a report was generated.		Pending
IT_015_07	Download Academic Records	Handle download interruption/failure	Report is generated; download failure can be simulated.	1. Start downloading. 2. Interrupt or simulate failure.	Generated report file	System reports the failure and allows a retry without presenting a corrupt file as successful.		Pending
ST_015_08	Download Academic Records	Check downloaded file opens and is readable	A report has been generated.	1. Download the report. 2. Open it using a compatible viewer.	Generated report	File opens successfully and its text/content is readable, with no empty or corrupted output.		Pending